

MONEY PUZZLE

Investing isn't a bet — it's a puzzle

Quant-based robo-advisor · Money Puzzle



Retail investors hit **three walls**

Information is scattered, decisions are gut feel, advice is out of reach

1

Hopping between apps

- Quotes in the broker app, news on portals, filings on DART
- Dozens of taps through tabs and menus to understand one stock
- And 'my portfolio' is never visible at a glance

2

All-in on one stock

- Picks come from paid chat rooms and word of mouth
- Buy and sell timing is gut feel
- Risk piles up in one place

3

Advice is distant and expensive

- Private banking serves mainly the wealthy
- Robo-advisors show results but never 'why'
- Number-heavy tables are hard for beginners

Scattered info · concentrated risk · distant advice — solved with one scroll

[illegible]

④ Filings · news ·
disclaimers

- A gesture everyone already knows
- The flow never breaks — each screen explains the next
- Korea · US · crypto on one line
- See → understand → buy, in one pass

Four apps — two to view, two to trade

Public

Money Puzzle Stock

Asset allocation by risk profile · Korea & US

Virtual portfolio · past returns

Real-time quotes

No order function

Public

Money Puzzle Coin

Crypto allocation by profile · spot only

Virtual portfolio · past returns

Real-time quotes

No order function

Founder pilot

Money Puzzle HTS

Live trading app linked to brokerage accounts

Basket trading of profile-matched portfolios

Monthly rebalancing

Demo open (/review/)

New · 2026.10

Money Puzzle Mini

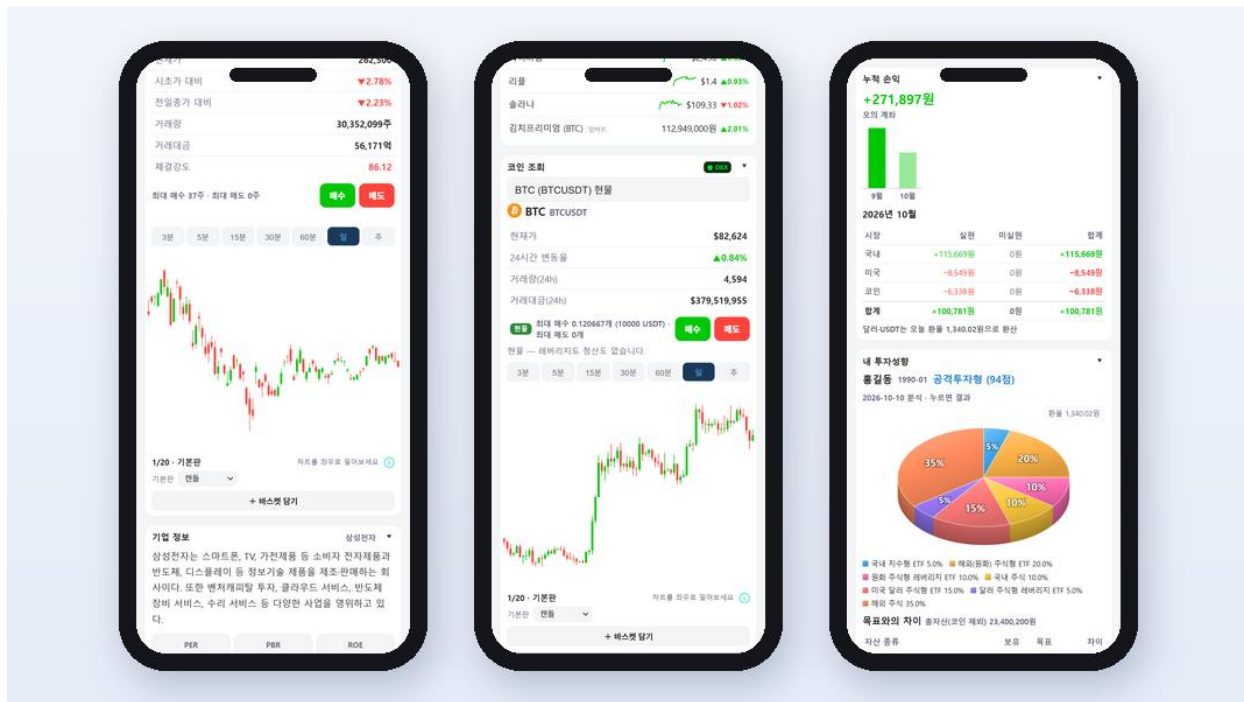
A light trading tool for people in their 20s–30s

Watchlist · charts

3-year income statement

No recommendations

One screen — stocks · crypto · my P&L



From left — stock lookup & chart · crypto lookup & chart · cumulative P&L and asset mix by profile

Three steps to my portfolio

Answer 7 questions → an asset mix that fits you, plus this month's picks

포트폴리오 선택

아래에서 하나를 고르십시오. 고른 포트폴리오에 맞추어 자산 종류별 비중과 종목을 보여 드립니다. 언제든 바꾸실 수 있습니다.

안정 포트폴리오 예금 또는 적금 수준의 수익률을 기대하며, 투자원금에 손실이 발생하는 것을 원하지 않는 유형

안정추구 포트폴리오 투자원금의 손실위험을 최소화하고, 이자소득이나 배당소득 수준의 안정적인 투자를 목표로 함. 다만, 수익을 위해 단기적인 손실을 수용할 수 있으며, 예·적금보다 높은 수익을 위해 자산 중 일부를 변동성 높은 상품에 투자할 의향이 있는 유형

✓ 위험중립 포트폴리오 원금손실, 고금리실현, 분배수익이, 파생상품, 환율, 인플레이션, 부동산 등 다양한 수익을 기대할 수 있으나 원금손실의 위험이 있는 유형

적극투자 포트폴리오 투자원금의 보전보다는 위험을 감내하더라도 높은 수준의 투자수익 실현을 추구함. 투자원금의 상당부분을 주식, 주식형펀드 또는 파생상품 등의 위험자산에 투자할 의향이 있는 유형

공격투자 포트폴리오 시장평균 수익률을 훨씬 넘어서는 높은 수준의 투자수익을 추구하며, 이를 위해 자산가치의 변동에 따른 손실 위험을 적극 수용. 투자자금 대부분을 주식, 주식형펀드 또는 파생상품 등의 위험자산에 투자할 의향이 있는 유형

1 Risk profile

KOFIA standard 7 questions → five levels, Conservative to Aggressive

추천 포트폴리오

위험중립 포트폴리오에 맞춘 자산 배분입니다. 아래 금액은 모든 이용자에 게 똑같이 주어지는 가상 금액입니다.

가상 계좌 100,000,000원

한도 1,355.28억



예금·MMF 30.0% 국내 채권형 ETF 15.0% 국내 원화 ETF 10.0% 해외 원화채시 채권형 ETF 10.0% 미국 달러 채권형 ETF 10.0% 미국 달러채시 채권형 ETF 10.0%

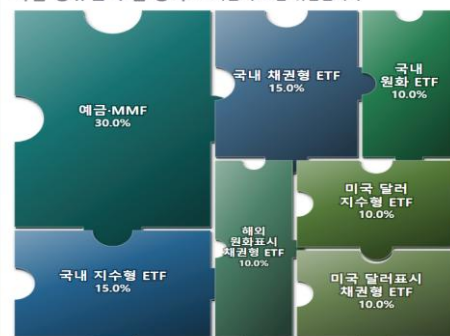
자산 종류	비율	금액
예금·MMF	30.0%	30,000,000원
국내 채권형 ETF	15.0%	15,000,000원
국내 원화 ETF	10.0%	10,000,000원
해외 원화채시 채권형 ETF	10.0%	10,000,000원
미국 달러 채권형 ETF	10.0%	10,000,000원
미국 달러채시 채권형 ETF	10.0%	10,000,000원

2 Asset allocation

Asset mix ratio by profile + picks per asset class (KR · US ETFs · stocks)

입니다.

자산 종류별 추천 종목 조각을 누르면 뒤집힙니다



3 Buy as a basket

Buy many names at once · rebalance on the 1st of every month

Quant-based robo-advisor

Data, not gut feel — pick with a proven momentum factor, trade by fixed rules

1 Blue chips only

- Top 50 by market cap, Korea & US
- ETFs in four groups: KRW/USD × plain/leveraged
- No inverse, ETNs or structured products

2 Uptrends only

- Only stocks above their 200-day average
- Downtrending stocks never make the ranking
- Stocks that closed limit-up the day before are excluded

3 Pick the 12-month leaders

- Rank by 12-month return, skipping the latest month
- 5 stocks · top 1 per ETF asset class
- Re-picked on the 1st of every month



Core strategy — Core + Satellite

Index ETFs (core) capture market return; 12-month momentum names (satellite) add excess return
6-year backtest · 0 of 100 random draws beat this rule · with auto-trading on, it's quant algorithmic investing

PROVEN RESULT

Side by side with the indices — **verified past returns**

If each app profile portfolio had started 1, 2, 3 years ago under today's engine rules

Term	Cons.	Mod. cons.	Moderate	Growth	Aggr.	KOSPI	KOSDAQ	S&P500	Nasdaq	Dow
1Y	+3.2%	+25.7%	+39.3%	+71.3%	+129.9%	+104.8%	+5.6%	+16.4%	+21.2%	+10.8%
2Y	+6.4%	+38.2%	+65.2%	+102.1%	+217.1%	+171.7%	+21.8%	+42.6%	+59.9%	+28.1%
3Y	+9.7%	+43.4%	+117.1%	+138.0%	+318.1%	+174.9%	+0.6%	+81.0%	+107.0%	+54.9%

Aggressive, 3 years: +318% — outperformed every index over the same period

Median of 20 start dates → 2026-10-02 · in KRW (US indices converted) · rebalanced monthly on the 1st · stock stop −15% · portfolio +30/−20% · idle cash, deposits, MMF at 3%/yr

KOSPI's 1–2Y surge was a concentrated market: SK hynix and Samsung Electronics made up 64–75% of the market-cap gain

When markets shook — we fell less

Largest drop over the same period (max drawdown)

Term	Cons.	Mod. cons.	Moderate	Growth	Aggr.	KOSPI	KOSDAQ	S&P500	Nasdaq	Dow
1Y	0.0%	-8.5%	-12.8%	-17.5%	-15.3%	-38.6%	-47.4%	-12.5%	-14.9%	-15.1%
2Y	0.0%	-8.5%	-13.3%	-17.5%	-18.4%	-38.6%	-47.4%	-17.8%	-23.1%	-17.0%
3Y	0.0%	-8.5%	-16.9%	-17.5%	-19.3%	-38.6%	-47.4%	-17.8%	-23.1%	-17.0%

※ KOSPI/KOSDAQ MDD comes from the 2026 sell-off inside the last year, so 1/2/3Y values match.

All five profiles

KOSPI -38.6% · KOSDAQ -47.4%
— all fell less than these

Crypto portfolio (BTC 50 · ETH 50) · last 1 year

App +9.4% Bitcoin -25.5%
 Max drawdown -10.2% vs Bitcoin -53.0%

Risk-adjusted too — twice the index

Not just return — 3 years by volatility (std. dev.) · return per unit of risk (Sharpe) · max drawdown

3Y	Cons.	Mod. cons.	Moderate	Growth	Aggr.	KOSPI	KOSDAQ	S&P500	Nasdaq	Dow
SR	—	1.23	1.68	1.47	1.88	1.03	0.08	0.96	1.00	0.72
Vol	0.0%	7.1%	13.5%	17.9%	24.5%	34.1%	33.0%	18.4%	23.0%	17.3%
MDD	0.0%	−8.5%	−16.9%	−17.5%	−19.3%	−38.6%	−47.4%	−17.8%	−23.1%	−17.0%

※ 3Y · median of 20 starts · Sharpe = (return − 3%) ÷ vol · β/α vs KOSPI 50 + S&P 500 50

Return per unit of risk (Sharpe) · 3Y

Aggressive 1.88

About 2× KOSPI 1.03 · S&P 500 0.96

Volatility & excess return vs market (Aggressive · 3Y)

β 1.00 α +19.4%/yr

Moves like the market · beats it by 19%p a year

What's different — how you see it and how it flows

	Typical broker app	Robo-advisor	Money Puzzle
Screen flow	Tabs · menus	Mostly a results screen	One vertical scroll
Unit of advice	Single stocks	Portfolio	Profile portfolio + stocks
Presentation	Number tables · charts	Pie chart	Puzzle · 3D tower · tree
Asset coverage	Mostly in-house products	Mostly in-house products	Korea · US · crypto on one screen
Evidence disclosed	—	Partial	Validation documented

Comparison is by typical service type

Market — already a big market in the US

	Wealthfront	Betterment	M1 Finance	Money Puzzle
AUM	About \$95B	\$70B+	About \$12B	Running virtual portfolios
Customers	1.4M+	1M+	1M+	—
Model	Automated allocation · rebalancing	Automated allocation · advisor platform (B2B)	Pie portfolios · auto-rebalancing	ETF core + quant satellites · Korea, US & crypto in one portfolio
Fees	0.25%/yr	[Unverified]	[Unverified]	[TBD]

Valuation What allocation-automation apps are worth — Wealthfront pursuing a Nasdaq listing (WLTH), \$8–10B valuation discussed

Sources: company filings · press (2025–2026) · amounts in USD ·

Revenue — customers · brokers · partners



B2C

Customer subscription

Paid personalized advice
Path via innovative financial service
(sandbox)



B2B

Broker PB tool license

A face-to-face advisory screen PBs use
with clients



Partnerships

Order-routing fee sharing

Partner broker accounts · order
integration
Basket buys · monthly rebalancing
orders

Value to brokers More orders and more deposits per customer → higher broker fee income (next two slides)

Higher broker fee income

The app itself is designed as a kind of sales machine.
Its trading style draws customers in and keeps deposits coming

1

Portfolio-level Basket buys · monthly rebalancing

- Buy many names as one basket, per the asset mix
- Auto-sell at stock stop or account-wide limits
- **All customers rebalance on the 1st · orders in bulk**
- **Deposits auto-bought toward target weights**

2

Crypto 12-slot strategy

- Profit sharing possible through partnerships with exchanges at home and abroad
- 12 consecutive days of buying spreads timing and lowers risk.
- More trading
- Spot trading only. No liquidations

e.g., one Aggressive customer

Up to 16 names per monthly rebalance (2 core ETFs + 14 picks)

12 times a year × 16 names = up to 192 rebalancing orders (buys)

Extra basket buys with every deposit · crypto bought daily in 12 slots

**More orders per
customer
= more fee income**

Higher broker deposits

Holding a profile portfolio at target weights takes real money.

The more aggressive the profile, the more names to hold — customers grow deposits on their own

Profile	Names	1 share of every name	Within 0.5%p of target	Name that sets the amount
Conservative	0	Not needed · leftover cash goes to MMF	—	—
Moderately conservative	1	≈ KRW 0.11M	≈ KRW 0.56M	KODEX 200 · target 20% · 1 share ≈ KRW 0.11M
Moderate	3	≈ KRW 1.17M	≈ KRW 4.12M	SPY · target 25% · 1 share ≈ KRW 1.03M
Growth	10	≈ KRW 3.82M	≈ KRW 131.42M	Samsung Electro-Mechanics · target 1.2% · 1 share ≈ KRW 1.58M
Aggressive	16	≈ KRW 10.95M	≈ KRW 236.55M	Samsung Electro-Mechanics · target 0.7% · 1 share ≈ KRW 1.58M

e.g., Aggressive — 1 share of all 16 ≈ KRW 10.95M · at target weights ≈ KRW 237M

Even short on cash, you buy 1 share,

→ the more you deposit, the closer to target weights · picks change monthly

※ 2026-10-02 pick list · closing prices, FX 1,346.8 KRW/USD

More deposits per customer
= **more deposits**

READY NOW

Not an idea — a service already running

4

Live apps

Stock · Coin · HTS · Mini

6 years

Backtests

Each rule tested one
change at a time

1

Patent filing

Chart gestures (page flip ·
range measure)

8

Design filings

Puzzle heatmap · 3D
tower · tree, etc.

2

Trademark filings

「머니퍼즐」 · 「MONEY
PUZZLE」

Running 24/7 on Naver Cloud · Try it — <https://emhwang3-boop.github.io/stock/>

4 apps today — next, brokers and the 20s–30s



Now

4 apps live

Stock · Coin · HTS · Mini
24/7 on Naver Cloud



Next

**Sandbox · outsourced
test**

Path to paid personalized
advice
Build a customer track record



Expand

Broker PB tool (B2B)

An advisory screen PBs use
with clients
Offered as a license



Broaden

Money Puzzle Mini

Expand to 20s–30s
Watchlist · charts · 3-year
income statement

Direction Build a track record with the consumer app (B2C) → expand to a broker PB tool (B2B)

The ask — Angel · Seed

Target KRW 100M–1B · KRW 100–300M / KRW 300M–1B

1 Stage · amount

- Stage — Angel · Seed
- Target — KRW 100–300M / KRW 300M–1B

2 Use of funds

- [People ○%]
- [Licensing · capital requirements ○%]
- [Servers · data licenses ○%]
- [Marketing ○%]

What we've built An engine verified by 6-year backtests · 4 live apps (Stock · Coin · HTS · Mini) · patent · design · trademark filings

Company · contact — Buman Inc.

Quant-based robo-advisor · Money Puzzle

1

Company

- Buman Inc.
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2

Contact

- Email — emhwang3@moneypuzzle.co.kr
- Website — emhwang3-boop.github.io/home/
- HTS demo — emhwang3-boop.github.io/review/

Developer [Fund-trading systems development, Investment Products, Citibank]

Investing isn't a bet — it's a puzzle

Scroll it like a webtoon — Money Puzzle

Thank you

① Yearly returns — not a single negative year

Profile portfolios vs indices · by year

	2023 (from start)	2024	2025	2026 (~10/2)
Conservative	+0.9%	+3.1%	+3.1%	+2.3%
Moderately conservative	+2.3%	+0.6%	+18.0%	+18.0%
Moderate	+4.8%	+19.5%	+33.2%	+30.3%
Growth	+4.8%	+17.4%	+41.7%	+36.6%
Aggressive	+8.0%	+35.6%	+32.5%	+115.5%
KOSPI	+4.2%	−9.6%	+75.6%	+66.2%
KOSDAQ	−2.4%	−21.7%	+36.5%	−3.5%
S&P500	+3.2%	+41.6%	+14.0%	+8.7%
Nasdaq	+5.5%	+47.7%	+17.9%	+12.7%
Dow	+5.2%	+29.6%	+10.7%	+2.5%

All five profiles positive in all four years · KOSPI −9.6% in 2024; KOSDAQ −2.4% in 2023, −21.7% in 2024, −3.5% in 2026

Median of 20 start dates 3 years ago (2023-09-05 to 10-02) · same yardstick as the 3Y returns on slide 8 · rebalanced monthly on the 1st · KRW (US indices converted) · deposits/MMF at 3%/yr

② What lifted KOSPI: chip concentration

What was behind KOSPI beating our portfolios (2025 · 2026)

Stock	2023 (9/25~)	2024	2025	2026 (~9/23)	3Y cumulative
SK hynix	+21%	+23%	+274%	+186%	+1,492%
Samsung Electronics	+13%	-32%	+125%	+139%	+313%
Samsung Electro-Mechanics	+11%	-19%	+106%	+489%	+985%
SK Square	+21%	+51%	+364%	+221%	+2,624%
KOSPI index	+6.4%	-9.6%	+75.6%	+68.0%	+183.7%
Of each year's total KOSPI market-cap gain, the share taken by Samsung Electronics and SK hynix	53%	—	47%	77%	64%

Why KOSPI led

- A few large caps surged — a cap-weighted index rides the names whose weight grows
- Our picks are equal-weighted — built to avoid concentration, so it lags in such markets and defends when large caps turn
- Two-thirds of Aggressive's Korean equity is a KODEX 200 core — it rides large-cap rallies too
- The portfolio is diversified — even Aggressive holds only 10% Korean stocks; the rest is US stocks and ETFs

3Y cumulative = compounded yearly returns · share based on the market-cap gain of the 788 Korean stocks in our data · 2024 excluded (total shrank)

③ Return vs volatility — Aggressive #1 over 3Y

Volatility · return ÷ volatility · positive months · worst month

	3Y volatility	Return ÷ vol, 1Y	2Y	3Y	Positive months (3Y)	Worst month (3Y)
Moderately conservative	7.1%	2.35	2.17	1.86	63%	−4.1%
Moderate	13.5%	2.14	2.24	2.22	68%	−10.5%
Growth	17.9%	2.79	2.52	2.24	63%	−12.4%
Aggressive	24.6%	3.62	2.88	2.57	71%	−14.6%
KOSPI	34.3%	1.94	1.62	1.21	58%	−22.2%
KOSDAQ	33.1%	0.12	0.22	0.06	55%	−21.4%
S&P500	18.4%	0.75	0.95	1.22	71%	−8.0%
Nasdaq	23.1%	0.76	1.04	1.20	68%	−10.8%
Dow	17.4%	0.48	0.67	0.92	66%	−7.6%

Volatility = daily return std. dev. × $\sqrt{252}$ · return ÷ vol = annual return ÷ annual vol (higher = more earned for the same swings) · Conservative excluded (deposit assumption only)

Each period started that day and run under today's engine rules · KRW · through 2026-10-02

④ How picks are chosen — better than random?

Aggressive (two-thirds core) · 1st of each month · annual return and MDD, changing only the pick method

Aggressive	Annual to 2025-12	Annual to 2026-10	MDD (full)
Adopted: 12-month return + stop −15 · portfolio +30/−20	+22.6%	+33.7%	−19.7%
12-month return only (no stop)	+24.7%	+34.8%	−38.9%
Index ETFs only (same weights)	+23.4%	+28.7%	−29.2%
100 random draws — median	+20.4%	+23.3%	—
Best of 100 random draws	+25.2%	+31.0%	—
Previous engine: Hull rule, 12 slots (old 09-25 weights)	+14.4%	+17.3%	−22.4%
12-month return 50% + 52-week high 50%	+19.1%	+23.2%	−29.3%
12-month return + low-vol slot split	+21.7%	+29.7%	−24.9%
Low-vol only	+18.6%	+19.5%	−18.7%
Just hold KODEX 200	+15.4%	+25.3%	−40.8%

From 2020-08-07 · 2026-10-02 asset mix · top 50 by cap · monthly switch · 5 stocks (above 200-day) · 1 per ETF class · 0.15% one-way · KRW · random = same pool, filters, count

12-month return = 12-month return skipping the latest month · beat all 100 random draws over the full period (0/100 · 1/100 through 2025)

⑤ Why these rules — date, exits, switching

Annual return and MDD, changing one setting at a time · adopted values marked (adopted)

Aggressive	~2025-12	~2026-10	MDD
1st of month (adopted)	+24.7%	+34.8%	−38.9%
8th of month	+25.8%	+33.0%	−42.3%
15th of month	+26.7%	+34.9%	−41.1%
22nd of month	+26.2%	+35.3%	−40.6%
Split into 4 staggered tranches	+25.9%	+34.5%	−40.7%
Stock take-profit +20	+21.4%	+26.9%	−19.6%
Stock take-profit +30	+19.6%	+24.7%	−19.2%
Stock take-profit +50	+22.2%	+29.0%	−20.2%
No stock take-profit (adopted)	+22.6%	+33.7%	−19.7%
Switch if outside top 5 (adopted)	+24.7%	+34.8%	−38.9%
Switch if outside top 7	+25.7%	+34.8%	−38.9%
Switch if outside top 10	+25.7%	+34.9%	−38.8%
Switch if outside top 15	+25.3%	+34.8%	−37.9%

Stop rules	~2025-12	~2026-10	MDD
Aggr. no rules	+24.7%	+34.8%	−38.9%
Aggr. stop −15	+21.7%	+32.5%	−28.3%
Aggr. PT −20	+24.7%	+34.8%	−38.9%
Aggr. adopted	+22.6%	+33.7%	−19.7%
Hold KODEX 200	+15.4%	+25.3%	−40.8%
Growth no rules	+18.7%	+24.6%	−28.2%
Growth stop −15	+18.5%	+25.2%	−23.3%
Growth adopted	+18.1%	+21.5%	−21.3%
Growth random	+19.0%	+23.5%	—
Growth index ETF	+19.1%	+24.2%	−25.2%

2020-08-07 – 2025-12-31 / – 2026-10-02 · new asset mix · top 50 by cap · first trading day · 0.15% one-way · KRW · idle cash 3% · date/switch rows without stops
Take-profit rows & adopted = stop −15 + portfolio +30/−20 (if hit, sell all, wait until next 1st)

⑥ How we measured — conditions

- Period — from the same date 1, 2, 3 years ago to 2026-10-02 (same as benchmark indices) · rule comparisons from 2020-08-07
- Portfolio — bought at the profile asset mix on the start date, rebalanced on the first trading day monthly · KRW account
- Core — part of the KR/US equity slots held in index ETFs (KODEX 200 · SPY): Aggressive two-thirds · Growth 40%
- Picks — top by 12-month return skipping the latest month · 5 stocks (above 200-day) · 1 per ETF class · index ETF if short
- Universe — top 50 by market cap in KOSPI/KOSDAQ · top 50 S&P 500 names by market cap (members on that day) · ETFs (no ETNs, ELS or other structured products; no inverse)
- Selling — stock stop −15% (intraday low) · no stock take-profit · sell all at portfolio +30% / −20% and wait until the 1st of next month
- Cash · costs — deposits, MMF and idle cash at 3%/yr · 0.15% one-way · taxes and FX fees excluded
- Indices — KOSPI, KOSDAQ, S&P 500, Nasdaq, Dow price indices (ex-dividends) · US indices converted to KRW (Yahoo KRW=X)
- Limits — 6 years of data with one big bear market (2022) · stocks dropped from the S&P 500 in between have no data and are missing (some survivorship bias) · ETF candidates are today's list
- Full source — 'Money Puzzle Investment Engine — Design Rationale and Validation Record v2.0' (confidential)